EXPERIMENT 12

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Outlier Detection Using IQR

Aim: Detect outliers in numerical data using the Interquartile Range method.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
data=np.array([10,12,13,15,16,18,19,20,22,100])
q1,q3=np.percentile(data,[25,75])
iqr=q3-q1
low=q1-1.5*iqr
high=q3+1.5*iqr
print("Outliers:",data[(data<low)|(data>high)])
clean=data[(data>=low)&(data<=high)]
print("Clean data:",clean)
plt.boxplot(data); plt.show()
```

Result

Values outside the lower and upper IQR limits are identified as outliers.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.